**F70LA LIFE INSURANCE MATHEMATICS - Sample solutions to 2024/25 December Exam**

1. (a)

$$\begin{aligned} {}_tp_x^\star &= \exp\left(-\int_0^t \mu_{x+s}^\star ds\right) \\ &= \exp\left(-\int_0^t 3\mu_{x+s} ds\right) \\ &= \left(\exp\left(-\int_0^t \mu_{x+s} ds\right)\right)^3 = \left({}_tp_x\right)^3. \end{aligned}$$

(b) (i) Under a constant force of mortality $\mu_x = 0.0125$, the Diplodocus' $t$-year survival probability from birth is ${}_tp_x = e^{-0.0125t}$ for $x \geq 0$.

The limiting age is $\infty$.

(ii) Letting the random variable $T_0$ represent the future lifetime of Diplodocus from age 0, its life expectancy from birth is

$$\mathbb{E}\left(T_0\right) = \int_0^\infty {}_tp_0 dt = \int_0^\infty e^{-0.0125t} dt = \frac{1}{0.0125} = 80 \text{ years}.$$

Similarly, the life expectancy from birth of T-rex is

$$\mathbb{E}\left(T_0^\star\right) = \frac{1}{3 \times 0.0125} = 26.667 \text{ years}.$$

Hence the difference in life expectancy is 53.3 years.

2. (a) Given

$$\ell_{80} = 10,000, \qquad p_{79} = 0.95 \qquad \text{and} \qquad q_{[x]} = 0.6q_x.$$

Since the select period is 1 year, $p_{[79]} = \frac{\ell_{80}}{\ell_{[79]}}$.

Also,

$$p_{[79]} = 1 - q_{[79]} = 1 - 0.6q_{79} = 1 - 0.6(1 - 0.95) = 0.97.$$

Hence

$$\ell_{[79]} = \frac{\ell_{80}}{p_{[79]}} = \frac{10,000}{0.97} = 10,309.278.$$

(b)

- A term assurance pays out if the insured life dies within the term of the policy.
- A risk to the insurance company is that lives die sooner and more often than anticipated, so the life office has to pay out more money sooner.
- Mortality rates used to price the term insurance are based on lives who have passed the required health checks.
- Therefore, their mortality rate is lower than that of the general population.
- Using select life tables means that premium rates more accurately reflect the mortality risk of the accepted lives,...
- ...Thus reducing the life office's risk.

3.

| $x$ | $\ell_x$ |
|---|---|
| 60 | 99,813 |
| 61 | 97,702 |
| 62 | 95,046 |

To apply a fractional age assumption, probabilities need to be considered within an integer age interval, i.e. between $[60, 61]$ and $[61, 62]$,

$$_{1.25}p_{60} = p_{60} \times {}_{0.25}p_{61}.$$

Then

$$p_{60} = \frac{\ell_{61}}{\ell_{60}} = \frac{97,702}{99,813} = 0.978850$$

and, using the UDD assumption,

$$_{0.25}p_{61} = \frac{\ell_{61.25}}{\ell_{61}} = \frac{0.75 \times \ell_{61} + 0.25 \times \ell_{62}}{\ell_{61}} = \frac{0.75 \times 97,702 + 0.25 \times 95,046}{97,702} = 0.993204.$$

Substituting for the calculated probabilities,

$$_{1.25}p_{60} = 0.978850 \times 0.993204 = 0.97220.$$

4. Using AM92 Ultimate and an interest rate of 6% per annum effective,

$$\begin{aligned}
\text{EPV} =& (50,000 - 1,000)\, A^{\,1}_{60:\overline{5}|} + 1,000\, (IA)^{\,1}_{60:\overline{5}|} \\
=& 49,000 \left(A_{60} - v^5\, {}_5p_{60}\, A_{65}\right) + 1,000 \left[(IA)_{60} - v^5\, {}_5p_{60}\, \left(5A_{65} + (IA)_{65}\right)\right] \\
=& 49,000\, [0.32692 - (0.70977)\, (0.40177)] + 1,000\, [5.46572 - (0.70977)\, (5(0.40177) + (5.50985))] \\
=& 2,046.089 + 129.172 \\
=& 2,175.26,
\end{aligned}$$

where

$$_5p_{60} = \frac{8821.2612}{9287.2164} = 0.94983 \quad \text{and} \quad v^5\, {}_5p_{60} = (1.06)^{-5}\, \frac{8821.2612}{9287.2164} = 0.70977.$$

Hence the single premium is £2,175.26.

5. (a)
- The annuity pays £1 per annum paid annually in advance.
- For the first $n$ years, regardless of whether $(x)$ is alive or not during that $n$-year period,...
- ...With the payment continuing past $n$ years only if $(x)$ survives $n$ years.
- The payment stops once $(x)$ dies.

(b)
- As the guaranteed term $n$ increases, the EPV increases...
- ...Because the EPV of providing £1 p.a. for certain in a given year is higher than the EPV of providing £1 p.a. conditional upon survival to the same year.

(c) Calculate, at $i = 4\%$ p.a, the expected present value

$$\ddot{a}_{\overline{10}|} + {}_{10}|\ddot{a}_{60}.$$

First term is

$$\ddot{a}_{\overline{10}|} = \frac{1 - 1.04^{-10}}{1 - 1.04^{-1}} = 8.43533.$$

Second term is

$$\begin{aligned} {}_{10}|\ddot{a}_{60} =& {}_{10}p_{60}\, v^{10}\, \ddot{a}_{70} \\ =& \frac{8054.0544}{9287.2164}\,(1.04)^{-10}\,(10.375) \\ =& 6.07832. \end{aligned}$$

Hence the expected present value is $8.43533 + 6.07832 = 14.5137$.

(d) The present value random variable $Y$ of the annuity is

$$Y = \begin{cases} \ddot{a}_{\overline{n}|} & \text{if } K_x \leq n-1 \\ \\ \ddot{a}_{\overline{K_x+1}|} & \text{if } K_x \geq n. \end{cases}$$

Alternative valid expressions gain full marks too, e.g. $Y = \ddot{a}_{\overline{n}|} + \max\{0, \ddot{a}_{\overline{K_x+1}|} - \ddot{a}_{\overline{n}|}\}$ or

$$Y = \ddot{a}_{\overline{n}|} + \begin{cases} 0 & \text{if } K_x \leq n-1 \\ \ddot{a}_{\overline{K_x+1}|} - \ddot{a}_{\overline{n}|} & \text{if } K_x \geq n, \end{cases}$$

| | |
|---|---|
| Mortality: | AM92 Select |
| Interest: | 6% per annum effective. |
| Expenses: | Initial: £250 plus 120% of the first monthly premium. |
| | Renewal: Nil. |
| | Claims: £300 at the time of payment of the sum assured. |

6.

(a) By the equivalence premium principle,

$$\text{EPV(Premiums)} = \text{EPV(Benefits)} + \text{EPV(Expenses)}.$$

Let $P$ be the annual premium.

- EPV(Premiums) is

$$P\ddot{a}_{[25]}^{(12)} \quad = 16.204P.$$

(Alternatively, Woolhouse's approximation gives $\ddot{a}_{[25]}^{(12)} = 1.0003\ddot{a}_{[25]} - 0.46812 = 16.199$.)

- EPV(Benefits) is

$$\begin{aligned} 100,000\,\bar{A}_{[25]} \quad &= 100,000\,\frac{0.06}{\ln(1.06)}\,(0.05686) \\ &= 100,000\,(0.05855) = 5,855.00. \end{aligned}$$

- EPV(Expenses) is

$$250 + 1.2\frac{P}{12} + 300\bar{A}_{[25]} \quad = 267.565 + 0.1P.$$

By the equivalence premium principle,

$$16.204P = 5,855.00 + 267.565 + 0.1P$$

Hence the monthly premium is $P/12 = 380.189/12 = £31.68$.

Mortality: AM92 Ultimate
Interest: 6% per annum effective.
Expenses: Renewal: Nil.
Claims: £500 at the time of payment of the sum assured.
Cost of the alteration: £600.

(b)

Let $S$ be the paid-up sum insured.

- The policy value just before the alteration is £10,995.
- Just after the alteration, the policy value is

$$\begin{aligned}\text{EPV(Benefits)} + \text{EPV(Expenses)} = (S+500)\,\bar{A}_{45} \quad &= (S+500)\,\frac{0.06}{\ln(1.06)}\,(0.15943) \\ &= £(S+500)\,(0.16417).\end{aligned}$$

- Cost of the alteration is £995.

The paid up sum insured satisfies

$$10,995 = 0.16417(S+500) + 995$$

| | |
|---|---|
| Mortality: | AM92 Ultimate. |
| Interest: | 4% per annum effective. |
| Expenses: | Initial: £200. |
| | Renewal: £30. |
| | Claims: Nil. |
| Bonus: | Simple reversionary bonus of 6% per annum vesting at the end of each policy year. |

7. (a) Three standard methods of calculating bonuses annually on a life insurance contract

- Simple bonus: the rate of bonus each year is a percentage of the initial (basic) sum assured under the policy.
- Compound bonus: the rate of bonus each year is a percentage of the initial (basic) sum assured and the bonuses added in the past.
- Super-compound bonus: two compound bonus rates are declared each year. The first rate is applied to the basic sum assured. The second rate is applied to the bonuses added to the policy in the past.

(b)

On death during the first year, the death benefit is the basic sum assured of £100,000. Due to the simple bonuses, the sum insured increases in each subsequent year by the fixed amount $6\% \times 100,000 = £6,000$. The pure endowment benefit amount is $£190,000 = 100,000 + 15 \times 6,000$. By the equivalence premium principle,

$$\text{EPV(Premiums)} = \text{EPV(Benefits)} + \text{EPV(Expenses)}$$

- EPV(Premiums) $= P\ddot{a}_{45:\overline{15}|} = 11.386P$.
- EPV(Benefits) $= 94,000\, A_{45:\overline{15}|} + 6,000\, (IA)^{1}_{45:\overline{15}|} + 96,000\, A_{45:\overline{15}|}^{1}$.
  Using AM92 Ultimate at 4% per annum effective,

$$A_{45:\overline{15}|} = 0.56206,$$

$$A_{45:\overline{15}|}^{1} = {}_{15}p_{45}\, v^{15} = \frac{9287.2164}{9801.3123}\,(1.04)^{-15} = 0.52614.$$

and

$$\begin{aligned}(IA)^{1}_{45:\overline{15}|} =& (IA)_{45} - {}_{15}p_{45}\, v^{15}\ ((IA)_{60} + 15A_{60}) \\ =& 8.33628 - (0.52614)\ (8.36234 + 15 \times 0.45640) \\ =& 0.33457\end{aligned}$$

Thus EPV(Benefits) $= 94,000\,(0.56206) + 6,000\,(0.33457) + 96,000\,(0.52614) = 105,350.50$.

- EPV(Expenses) $= 200 + 30\left(\ddot{a}_{45:\overline{15}|} - 1\right) = 200 + 30 \times 10.386 = 511.58$.

Hence from EPV(Premiums) = EPV(Benefits) + EPV(Expenses),

$$11.386P = 105,350.50 + 511.58 \qquad \Rightarrow \qquad P = \frac{105,862.08}{11.386} = £9,297.57.$$

(c)
- The expected cost of the benefits and expenses is spread out among all future premium payments, i.e. only a portion of the cost is included in each premium.
- If the insured life died in 2 years' time, the insurer would only have collected 2 years' worth of premiums.
- The insurer would not have collected via premiums the value of the benefits paid out.
- Therefore, the insurer would make a loss.